

1. INTRODUCTION TO COMPUTER

About this chapter

Computers are an integral part of our lives. Wherever we are; sitting in our homes, working in the office, driving on roads, staying in a hotel, etc. — our lives are directly or indirectly affected by the computers. In this era of information, we are dependent on the storage, flow and processing of data and information, which can only be possible with the help of computers. The purpose of this chapter is to introduce you to the “computer”.

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1.1 INTRODUCTION

Nowadays, [computers are an integral part of our lives](#). They are used for the reservation of tickets for airplanes and railways, payment of telephone and electricity bills, deposit and withdrawal of money from banks, processing of business data, forecasting of weather conditions, diagnosis of diseases, searching for information on the Internet, etc. Computers are also used extensively in schools, universities, organizations, music industry, movie industry, scientific research, law firms, fashion industry, etc.

[The term computer is derived from the word compute](#). The word *compute* means *to calculate*. A *computer* is an electronic machine that accepts data from the user, processes the data by performing calculations and operations on it, and generates the desired output results. Computer performs both simple and complex operations, with speed and accuracy.

This chapter discusses the [history and evolution of computer](#), the concept of input-process-output and the characteristics of computer. This chapter also discusses the classification of digital computers based on their size and type, and the application of computer in different domain areas.

1.2 DIGITAL AND ANALOG COMPUTERS

A [digital computer](#) uses distinct values to represent the data internally. All information are represented using the digits 0s and 1s. The computers that we use at our homes and offices are digital computers.

[Analog computer](#) is another kind of a computer that represents data as variable across a continuous range of values. The earliest computers were analog computers. Analog computers are used for measuring of parameters that vary continuously in real time, such as temperature, pressure and voltage. Analog computers may be more flexible but generally less precise than digital computers. Slide rule is an example of an analog computer.

This book deals only with the *digital computer* and uses the term *computer* for them.

1.3 CHARACTERISTICS OF COMPUTER

Speed, accuracy, diligence, storage capability and versatility are some of the key characteristics of a computer. A brief overview of these characteristics are—

- **Speed** The computer can process data very fast, at the rate of millions of instructions per second. Some calculations that would have taken hours and days to complete otherwise, can be completed in a few seconds using the computer. For example, calculation and generation of salary slips of thousands of employees of an organization, weather forecasting that requires analysis of a large amount of data related to temperature, pressure and humidity of various places, etc.

- **Accuracy** Computer provides a high degree of accuracy. For example, the computer can accurately give the result of division of any two numbers up to 10 decimal places.
- **Diligence** When used for a longer period of time, the computer does not get tired or fatigued. It can perform long and complex calculations with the same speed and accuracy from the start till the end.
- **Storage Capability** Large volumes of data and information can be stored in the computer and also retrieved whenever required. A limited amount of data can be stored, temporarily, in the primary memory. Secondary storage devices like floppy disk and compact disk can store a large amount of data permanently.
- **Versatility** Computer is versatile in nature. It can perform different types of tasks with the same ease. At one moment you can use the computer to prepare a letter document and in the next moment you may play music or print a document.

Computers have several limitations too.

- Computer can only perform tasks that it has been programmed to do.
- Computer cannot do any work without instructions from the user.
- It executes instructions as specified by the user and does not take its own decisions.

1.4 HISTORY OF COMPUTER

Until the development of the first generation computers based on vacuum tubes, there had been several developments in the computing technology related to the mechanical computing devices. The key developments that took place till the first computer was developed are as follows—

- **Calculating Machines** ABACUS was the first mechanical calculating device for counting of large numbers. The word ABACUS means calculating board. It consists of bars in horizontal positions on which sets of beads are inserted. The horizontal bars have 10 beads each, representing units, tens, hundreds, etc.



Figure 1.1 Abacus

- **Napier's Bones** was a mechanical device built for the purpose of multiplication in 1617 ad. by an English mathematician John Napier.
- **Slide Rule** was developed by an English mathematician Edmund Gunter in the 16th century. Using the slide rule, one could perform operations like addition, subtraction,

multiplication and division. It was used extensively till late 1970s.

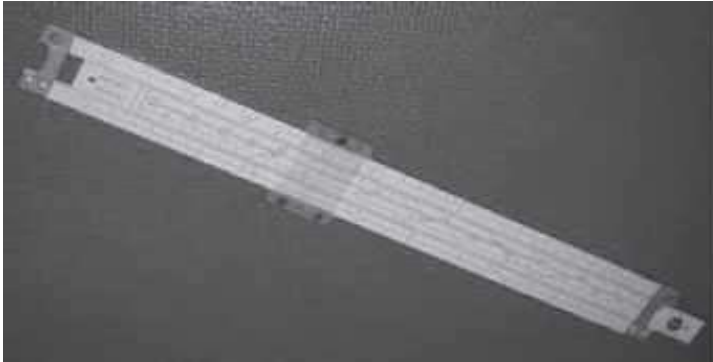


Figure 1.2 Slide rule

- ***Pascal's Adding and Subtraction Machine*** was developed by Blaise Pascal. It could add and subtract. The machine consisted of wheels, gears and cylinders.
- ***Leibniz's Multiplication and Dividing Machine*** was a mechanical device that could both multiply and divide. The German philosopher and mathematician Gottfried Leibniz built it around 1673.
- ***Punch Card System*** was developed by Jacquard to control the power loom in 1801. He invented the punched card reader that could recognize the presence of hole in the punched card as binary one and the absence of the hole as binary zero. [The 0s and 1s are the basis of the modern digital computer.](#)



Figure 1.3 Punched card

- ***Babbage's Analytical Engine*** An English man Charles Babbage built a mechanical machine to do complex mathematical calculations, in the year 1823. The machine was called as [difference engine](#). Later, Charles Babbage and Lady Ada Lovelace developed a general-purpose calculating machine, the [analytical engine](#). Charles Babbage is also [called the father of computer](#).
- ***Hollerith's Punched Card Tabulating Machine*** was invented by Herman Hollerith. The [machine could read the information from a punched card and process it electronically.](#)

The developments discussed above and several others not discussed here, resulted in the development of the first computer in the 1940s.

1.5 GENERATIONS OF COMPUTER

The computer has evolved from a large—sized simple calculating machine to a smaller but much more powerful machine. The evolution of computer to the current state is defined in terms of the generations of computer. Each generation of computer is designed based on a new technological development, resulting in better, cheaper and smaller computers that are more powerful, faster and efficient than their predecessors. Currently, there are five generations of computer. In the following subsections, we will discuss the generations of computer in terms of—

1. the technology used by them (hardware and software),
2. computing characteristics (speed, i.e., number of instructions executed per second),
3. physical appearance, and
4. their applications.

1.5.1 First Generation (1940 to 1956): Using Vacuum Tubes

- **Hardware Technology** The first generation of computers used **vacuum tubes** for circuitry and **magnetic drums for memory**. The input to the computer was through punched cards and paper tapes. The output was displayed as printouts.

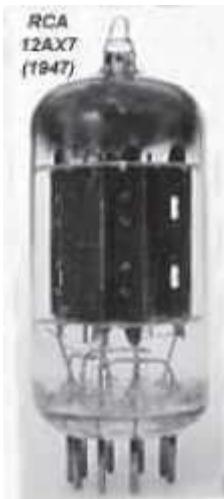


Figure 1.4 Vacuum tube

- **Software Technology** The instructions were written in **machine language**. Machine language uses 0s and 1s for coding of the instructions. The first generation computers could solve one problem at a time.
- **Computing Characteristics** The computation time was in **milliseconds**.
- **Physical Appearance** These computers were enormous in size and required a large room for installation.
- **Application** They were used for scientific applications as they were the fastest computing device of their time.
- **Examples** UNIVersal Automatic Computer (UNIVAC), Electronic Numerical Integrator And Calculator (ENIAC), and Electronic Discrete Variable Automatic Computer (EDVAC).

The first generation computers used a large number of vacuum tubes and thus generated a lot of

heat. They consumed a great deal of electricity and were expensive to operate. The machines were prone to frequent malfunctioning and required constant maintenance. Since first generation computers used machine language, they were difficult to program.

1.5.2 Second Generation (1956 to 1963): Using Transistors

- **Hardware Technology** Transistors replaced the vacuum tubes of the first generation of computers. Transistors allowed computers to become smaller, faster, cheaper, energy efficient and reliable. The second generation computers used *magnetic core technology* for primary memory. They used magnetic tapes and magnetic disks for secondary storage. The input was still through punched cards and the output using printouts. They used the concept of a stored program, where instructions were stored in the memory of computer.



Figure 1.5 Transistors

- **Software Technology** The instructions were written using the *assembly language*. **Assembly language uses mnemonics like ADD for addition and SUB for subtraction** for coding of the instructions. It is easier to write instructions in assembly language, as compared to writing instructions in machine language. High-level programming languages, such as early versions of COBOL and FORTRAN were also developed during this period.
- **Computing Characteristics** The computation time was in **microseconds**.
- **Physical Appearance** Transistors are smaller in size compared to vacuum tubes, thus, the size of the computer was also reduced.
- **Application** The cost of commercial production of these computers was very high, though less than the first generation computers. The transistors had to be assembled manually in second generation computers.
- **Examples** PDP-8, IBM 1401 and CDC 1604.

Second generation computers generated a lot of heat but much less than the first generation computers. They required less maintenance than the first generation computers.

1.5.3 Third Generation (1964 to 1971): Using Integrated Circuits

- **Hardware Technology** The third generation computers used the *Integrated Circuit (IC)* chips. In an IC chip, multiple transistors are placed on a silicon chip. Silicon is a type of semiconductor. The use of IC chip increased the speed and the efficiency of computer, manifold. The **keyboard and monitor were used** to interact with the third generation computer, instead of the punched card and printouts.



Figure 1.6 IC chips

- **Software Technology** The keyboard and the monitor were interfaced through the *operating system*. Operating system allowed different applications to run at the same time. **High-level languages** were used extensively for programming, instead of machine language and assembly language.
- **Computing Characteristics** The computation time was in **nanoseconds**.
- **Physical Appearance** The size of these computers was quite small compared to the second generation computers.
- **Application** Computers became accessible to mass audience. Computers were produced commercially, and were smaller and cheaper than their predecessors.
- **Examples** IBM 370, PDP 11.

The third generation computers used less power and generated less heat than the second generation computers. The cost of the computer reduced significantly, as individual components of the computer were not required to be assembled manually. The maintenance cost of the computers was also less compared to their predecessors.

1.5.4 Fourth Generation (1971 to present): Using Microprocessors

- **Hardware Technology** They use the **Large Scale Integration (LSI)** and the *Very Large Scale Integration (VLSI)* technology. Thousands of transistors are integrated on a small silicon chip using LSI technology. VLSI allows hundreds of thousands of components to be integrated in a small chip. This era is marked by the development of **microprocessor**. *Microprocessor* is a chip containing millions of transistors and components, and,

designed using LSI and VLSI technology. A microprocessor chip is shown in Figure 1.7. This generation of computers gave rise to Personal Computer (PC). [Semiconductor memory replaced the earlier magnetic core memory, resulting in fast random access](#) to memory. Secondary storage device like magnetic disks became smaller in physical size and larger in capacity. The [linking of computers](#) is another key development of this era. The computers were linked to form networks that led to the emergence of the Internet. This generation also saw the development of [pointing devices like mouse](#), and handheld devices.

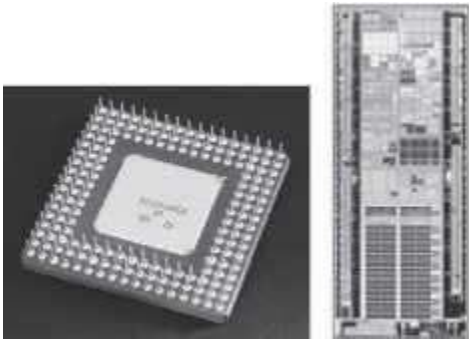


Figure 1.7 Microprocessors

- **Software Technology** Several new operating systems like the MS-DOS and MS-Windows developed during this time. This generation of computers supported *Graphical User Interface (GUI)*. GUI is a user-friendly interface that allows user to interact with the computer via menus and icons. High-level programming languages are used for the writing of programs.
- **Computing Characteristics** The computation time is in [picoseconds](#).
- **Physical Appearance** They are smaller than the computers of the previous generation. Some can even fit into the palm of the hand.
- **Application** They became widely available for commercial purposes. Personal computers became available to the home user.
- **Examples** The Intel 4004 chip was the first microprocessor. The components of the computer like Central Processing Unit (CPU) and memory were located on a single chip. In 1981, IBM introduced the first computer for home use. In 1984, Apple introduced the Macintosh.

The microprocessor has resulted in the fourth generation computers being smaller and cheaper than their predecessors. The fourth generation computers are also portable and more reliable. They generate much lesser heat and require less maintenance compared to their predecessors. GUI and pointing devices facilitate easy use and learning on the computer. Networking has resulted in resource sharing and communication among different computers.

1.5.5 Fifth Generation (Present and Next): Using Artificial Intelligence

The goal of fifth generation computing is to develop computers that are capable of learning and self-organization. The fifth generation computers use [Super Large Scale Integrated \(SLSI\)](#) chips

that are able to store millions of components on a single chip. These computers have large memory requirements.

This generation of computers uses *parallel processing* that allows several instructions to be executed in parallel, instead of serial execution. Parallel processing results in faster processing speed. The Intel dual core microprocessor uses parallel processing.

The fifth generation computers are based on *Artificial Intelligence (AI)*. They try to simulate the human way of thinking and reasoning. Artificial Intelligence includes areas like Expert System (ES), Natural Language Processing (NLP), speech recognition, voice recognition, robotics, etc.

1.6 CLASSIFICATION OF COMPUTER

The digital computers that are available nowadays vary in their sizes and types. The computers are broadly classified into four categories (Figure 1.8) based on their size and type—(1) Microcomputers, (2) Minicomputers, (3) Mainframe computers, and (4) Supercomputer.

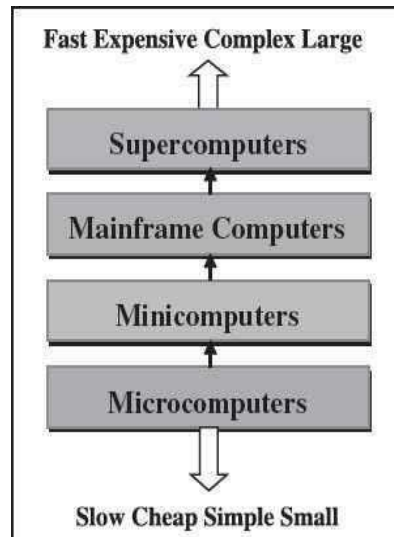


Figure 1.8 Classification of computers based on size and type

1.6.1 Microcomputers

Microcomputers are **small, low-cost and single-user digital computer**. They consist of CPU, input unit, output unit, storage unit and the software. Although microcomputers are stand-alone machines, they can be connected together to create a network of computers that can serve more than one user. IBM PC based on Pentium microprocessor and Apple Macintosh are some

examples of microcomputers. Microcomputers include desktop computers, notebook computers or laptop, tablet computer, handheld computer, smart phones and netbook, as shown in Figure 1.9.



Figure 1.9 Microcomputers

- **Desktop Computer or Personal Computer (PC)** is the most common type of microcomputer. It is a stand-alone machine that can be placed on the desk. Externally, it consists of three units—keyboard, monitor, and a system unit containing the CPU, memory, hard disk drive, etc. It is not very expensive and is suited to the needs of a single user at home, small business units, and organizations. Apple, Microsoft, HP, Dell and Lenovo are some of the PC manufacturers.
- **Notebook Computers or Laptop** resemble a notebook. They are portable and have all the features of a desktop computer. The advantage of the laptop is that it is small in size (can be put inside a briefcase), can be carried anywhere, has a battery backup and has all the functionality of the desktop. Laptops can be placed on the lap while working (hence the name). Laptops are costlier than the desktop machines.
- **Netbook** These are smaller notebooks optimized for low weight and low cost, and are designed for accessing web-based applications. Starting with the earliest netbook in late 2007, they have gained significant popularity now. Netbooks deliver the performance needed to enjoy popular activities like streaming videos or music, emailing, Web surfing or instant messaging. The word *netbook* was created as a blend of Internet and notebook.

- **Tablet Computer** has features of the notebook computer but it can accept input from a stylus or a pen instead of the keyboard or mouse. It is a portable computer. Tablet computer are the new kind of PCs.
- **Handheld Computer or Personal Digital Assistant (PDA)** is a small computer that can be held on the top of the palm. It is small in size. Instead of the keyboard, PDA uses a pen or a stylus for input. PDAs do not have a disk drive. They have a limited memory and are less powerful. PDAs can be connected to the Internet via a wireless connection. Casio and Apple are some of the manufacturers of PDA. Over the last few years, PDAs have merged into mobile phones to create smart phones.
- **Smart Phones** are cellular phones that function both as a phone and as a small PC. They may use a stylus or a pen, or may have a small keyboard. They can be connected to the Internet wirelessly. They are used to access the electronic-mail, download music, play games, etc. Blackberry, Apple, HTC, Nokia and LG are some of the manufacturers of smart phones.

1.6.2 Minicomputers

Minicomputers are digital computers, generally used in multi-user systems. They have high processing speed and high storage capacity than the microcomputers. Minicomputers can support 4–200 users simultaneously. The users can access the minicomputer through their PCs or terminal. They are used for real-time applications in industries, research centers, etc. PDP 11, IBM (8000 series) are some of the widely used minicomputers.



Figure 1.10 Minicomputer

1.6.3 Mainframe Computers

Mainframe computers are multi-user, multi-programming and high performance computers. They operate at a very high speed, have very large storage capacity and can handle the workload of many users. Mainframe computers are large and powerful systems generally used in centralized databases. The user accesses the mainframe computer via a terminal that may be a dumb terminal, an intelligent terminal or a PC. A *dumb terminal* cannot store data or do processing of its own. It has the input and output device only. An *intelligent terminal* has the input and output device, can do processing, but, cannot store data of its own. The dumb and the intelligent terminal use the processing power and the storage facility of the mainframe computer. Mainframe computers are used in organizations like banks or companies, where many people

require frequent access to the same data. Some examples of mainframes are CDC 6600 and IBM ES000 series.



Figure 1.11 Mainframe computer

1.6.4 Supercomputers

Supercomputers are the fastest and the most expensive machines. They have high processing speed compared to other computers. The speed of a supercomputer is generally measured in FLOPS (FLoating point Operations Per Second). Some of the faster supercomputers can perform trillions of calculations per second. [Supercomputers are built by interconnecting thousands of processors that can work in parallel.](#)

Supercomputers are used for highly calculation-intensive tasks, such as, weather forecasting, climate research (global warming), molecular research, biological research, nuclear research and aircraft design. They are also used in major universities, military agencies and scientific research laboratories. Some examples of supercomputers are IBM Roadrunner, IBM Blue gene and Intel ASCI red. PARAM is a series of supercomputer assembled in India by C-DAC (Center for Development of Advanced Computing), in Pune. PARAM Padma is the latest machine in this series. The peak computing power of PARAM Padma is 1 Tera FLOP (TFLOP).



Figure 1.12 Supercomputer

1.7 THE COMPUTER SYSTEM

Computer is an electronic device that accepts data as input, processes the input data by [performing mathematical and logical operations on it](#), and gives the desired output. The computer system consists of four parts•(1) Hardware, (2) Software, (3) Data, and (4) Users. The parts of computer system are shown in [Figure 1.13](#).

Hardware consists of the mechanical parts that make up the computer as a machine. The hardware consists of physical devices of the computer. The devices are required for input, output, storage and processing of the data. Keyboard, monitor, hard disk drive, floppy disk drive, printer, processor and motherboard are some of the hardware devices.

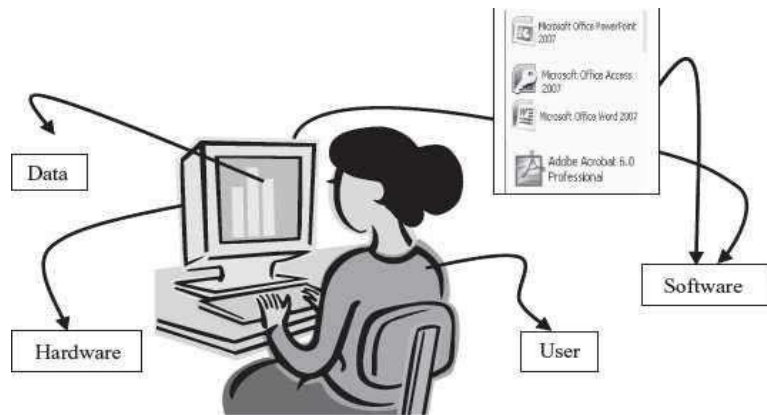


Figure 1.13 Parts of computer system

Software is a set of instructions that tells the computer about the tasks to be performed and how these tasks are to be performed. *Program* is a set of instructions, written in a language understood by the computer, to perform a specific task. [A set of programs and documents are collectively called software.](#) The hardware of the computer system cannot perform any task on its own. The hardware needs to be instructed about the task to be performed. Software instructs the computer about the task to be performed. The hardware carries out these tasks. Different software can be loaded on the same hardware to perform different kinds of tasks.

Data are isolated values or raw facts, which by themselves have no much significance. For example, the data like 29, *January*, and 1994 just represent values. The data is provided as input to the computer, which is processed to generate some meaningful information. For example, 29, January and 1994 are processed by the computer to give the date of birth of a person.

Users are [people who write computer programs or interact with the computer.](#) They are also known as *skinware*, *liveware*, *humanware* or *peopleware*. Programmers, data entry operators, system analyst and computer hardware engineers fall into this category.

1.7.1 The Input-Process-Output Concept

A computer is an electronic device that (1) accepts data, (2) processes data, (3) generates output, and (4) stores data. The concept of generating output information from the input data is also referred to as *input-process-output* concept.



The input-process-output concept of the computer is explained as follows—

- **Input** The computer accepts input data from the user via an input device like keyboard. The input data can be characters, word, text, sound, images, document, etc.
- **Process** The computer processes the input data. For this, it performs some actions on the data by using the instructions or program given by the user of the data. The action could be an arithmetic or logic calculation, editing, modifying a document, etc. During processing, the data, instructions and the output are stored temporarily in the computer's main memory.
- **Output** The output is the result generated after the processing of data. The output may be in the form of text, sound, image, document, etc. The computer may display the output on a monitor, send output to the printer for printing, play the output, etc.
- **Storage** The input data, instructions and output are stored permanently in the secondary storage devices like disk or tape. The stored data can be retrieved later, whenever needed.

1.7.2 Components of Computer Hardware

The computer system hardware comprises of three main components —

1. Input/Output (I/O) Unit,
2. Central Processing Unit (CPU), and
3. Memory Unit.

The I/O unit consists of the input unit and the output unit. CPU performs calculations and processing on the input data, to generate the output. The memory unit is used to store the data, the instructions and the output information. [Figure 1.14](#) illustrates the typical interaction among the different components of the computer.

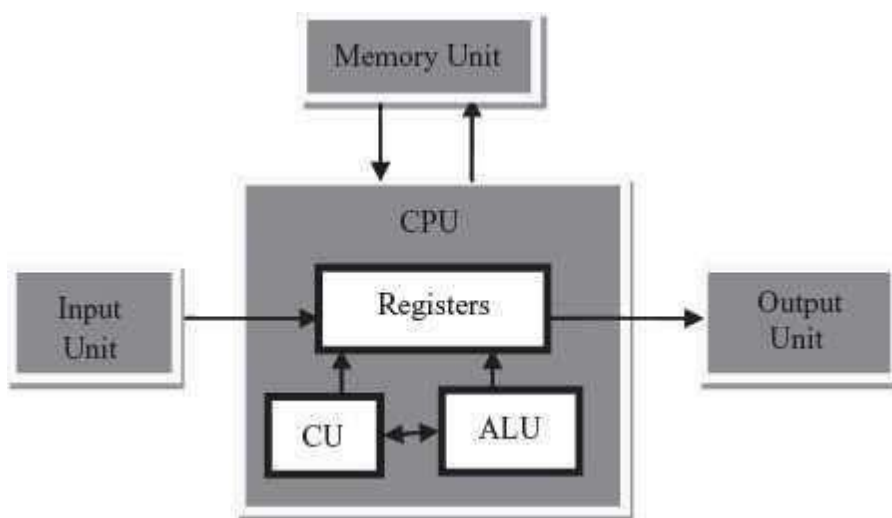


Figure 1.14 The computer system interaction

- **Input/Output Unit** The user interacts with the computer via the I/O unit. The Input unit accepts data from the user and the Output unit provides the processed data i.e. the information to the user. The Input unit converts the data that it accepts from the user, into a form that is understandable by the computer. Similarly, the Output unit provides the output in a form that is understandable by the user. The input is provided to the computer [using input devices like keyboard, trackball and mouse](#). Some of the commonly used output devices are [monitor and printer](#).
- **Central Processing Unit** CPU controls, coordinates and supervises the operations of the computer. It is responsible for processing of the input data. CPU consists of Arithmetic Logic Unit (ALU) and Control Unit (CU).
 - ALU performs all the arithmetic and logic operations on the input data.
 - CU controls the overall operations of the computer i.e. it checks the sequence of execution of instructions, and, controls and coordinates the overall functioning of the units of computer.

Additionally, CPU also has a set of *registers* for temporary storage of data, instructions, addresses and intermediate results of calculation.

- **Memory Unit** Memory unit stores the data, instructions, intermediate results and output, *temporarily*, during the processing of data. This memory is also called the [main memory or primary memory](#) of the computer. The input data that is to be processed is brought into the main memory before processing. The instructions required for processing of data and any intermediate results are also stored in the main memory. The output is stored in memory before being transferred to the output device. CPU can work with the information stored in the main memory. Another kind of storage unit is also referred to as the [secondary memory](#) of the computer. The data, the programs and the output are stored *permanently* in the storage unit of the computer. [Magnetic disks, optical disks and magnetic tapes are examples of secondary memory](#).

1.8 APPLICATION OF COMPUTERS

Computers have proliferated into various areas of our lives. For a user, computer is a tool that provides the desired information, whenever needed. You may use computer to get information about the reservation of tickets (railways, airplanes and cinema halls), books in a library, medical history of a person, a place in a map, or the dictionary meaning of a word. The information may be presented to you in the form of text, images, video clips, etc.

[Figure 1.15](#) shows some of the applications of computer. Some of the application areas of the computer are listed below—

- **Education** Computers are extensively used, as a tool and as an aid, for imparting education. Educators use computers to prepare notes and presentations of their lectures. Computers are used to develop computer-based training packages, to provide distance education using the e-learning software, and to conduct online examinations. Researchers use computers to get easy access to conference and journal details and to get global access to the research material.

- **Entertainment** Computers have had a major impact on the entertainment industry. The user can download and view movies, play games, chat, book tickets for cinema halls, use multimedia for making movies, incorporate visual and sound effects using computers, etc. The users can also listen to music, download and share music, create music using computers, etc.
- **Sports** A computer can be used to watch a game, view the scores, improve the game, play games (like chess, etc.) and create games. They are also used for the purposes of training players.
- **Advertising** Computer is a powerful advertising media. Advertisement can be displayed on different websites, electronic-mails can be sent and reviews of a product by different customers can be posted. Computers are also used to create an advertisement using the visual and the sound effects. For the advertisers, computer is a medium via which the advertisements can be viewed globally. Web advertising has become a significant factor in the marketing plans of almost all companies. In fact, [the business model of Google is mainly dependent on web advertising for generating revenues.](#)

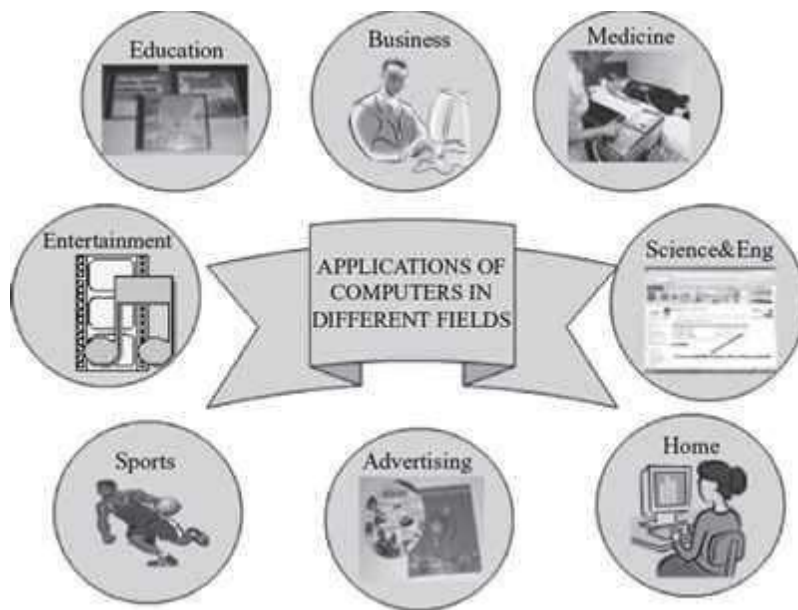


Figure 1.15 Applications of computer

- **Medicine** Medical researchers and practitioners use computers to access information about the advances in medical research or to take opinion of doctors globally. The medical history of patients is stored in the computers. Computers are also an integral part of various kinds of sophisticated medical equipment like ultrasound machine, CAT scan machine, MRI scan machine, etc. Computers also provide assistance to the medical surgeons during critical surgery operations like laparoscopic operations, etc.
- **Science and Engineering** Scientists and engineers use computers for performing complex scientific calculations, for designing and making drawings (CAD/CAM applications) and also for simulating and testing the designs. Computers are used for storing the complex data, performing complex calculations and for visualizing 3–

dimensional objects. Complex scientific applications like the launch of the rockets, space exploration, etc., are not possible without the computers.

- **Government** The government uses computers to manage its own operations and also for e-governance. The websites of the different government departments provide information to the users. Computers are used for the filing of income tax return, paying taxes, online submission of water and electricity bills, for the access of land record details, etc. The police department uses computers to search for criminals using fingerprint matching, etc.
- **Home** Computers have now become an integral part of home equipment. At home, people use computers to play games, to maintain the home accounts, for communicating with friends and relatives via Internet, for paying bills, for education and learning, etc. Microprocessors are embedded in house hold utilities like, washing machines, TVs, food processors, home theatres, security devices, etc.

The list of applications of computers is so long that it is not possible to discuss all of them here.

In addition to the applications of the computers discussed above, computers have also proliferated into areas like [banks](#), [investments](#), [stock trading](#), [accounting](#), [ticket reservation](#), [military operations](#), [meteorological predictions](#), [social networking](#), [business organizations](#), [police department](#), [video conferencing](#), [telepresence](#), [book publishing](#), [web newspapers](#), and [information sharing](#).

SUMMARY

- *Computer* is an electronic device which accepts data as input, performs processing on the data, and gives the desired output. A computer may be *analog or digital computer*.
- Speed, accuracy, diligence, storage capability and versatility are the main *characteristics of computer*.
- The *computing devices* have evolved from simple mechanical machines, like ABACUS, Napier's bones, Slide Rule, Pascal's Adding and Subtraction Machine, Leibniz's Multiplication and Dividing Machine, Jacquard Punched Card System, Babbage's Analytical Engine and Hollerith's Tabulating Machine, to the first electronic computer.
- Charles Babbage is called the father of computer.
- The evolution of computers to their present state is divided into *five generations of computers*, based on the hardware and software they use, their physical appearance and their computing characteristics.
- *First generation computers* were vacuum tubes based machines. These were large in size, expensive to operate and instructions were written in machine language. Their computation time was in milliseconds.
- *Second generation computers* were transistor based machines. They used the stored program concept. Programs were written in assembly language. They were smaller in size, less expensive and required less maintenance than the first generation computers. The computation time was in microseconds.
- *Third generation computers* were characterized by the use of IC. They consumed less power and required low maintenance compared to their predecessors. High-level languages were used for programming. The computation time was in nanoseconds. These

computers were produced commercially.

- *Fourth generation computers* used microprocessors which were designed using the LSI and VLSI technology. The computers became small, portable, reliable and cheap. The computation time is in picoseconds. They became available both to the home user and for commercial use.
- *Fifth generation computers* are capable of learning and self organization. These computers use SLSI chips and have large memory requirements. They use parallel processing and are based on AI. The fifth generation computers are still being developed.
- *Computers are broadly classified* as microcomputers, minicomputers, mainframe computers, and supercomputers, based on their sizes and types.
- *Microcomputers* are small, low-cost standalone machines. Microcomputers include desktop computers, notebook computers or laptop, netbooks, tablet computer, handheld computer and smart phones.
- *Minicomputers* are high processing speed machines having more storage capacity than the microcomputers. Minicomputers can support 4–200 users simultaneously.
- *Mainframe computers* are multi-user, multiprogramming and high performance computers. They have very high speed, very large storage capacity and can handle large workloads. Mainframe computers are generally used in centralized databases.
- *Supercomputers* are the most expensive machines, having high processing speed capable of performing trillions of calculations per second. The speed of a supercomputer is measured in FLOPS. Supercomputers find applications in computing-intensive tasks.
- *Computer* is an electronic device based on the input-process-output concept. Input/Output Unit, CPU and Memory unit are the three main *components of computer*.
- *Input/Output Unit* consists of the Input unit which accepts data from the user and the Output unit that provides the processed data. *CPU* processes the input data, and, controls, coordinates and supervises the operations of the computer. CPU consists of ALU, CU and Registers. The memory unit stores programs, data and output, temporarily, during the processing. Additionally, storage unit or secondary memory is used for the storing of programs, data and output permanently.
- Computers are used in various areas of our life. Education, entertainment, sports, advertising, medicine, science and engineering, government, office and home are some of the *application areas of the computers*.

KEYWORDS

ABACUS	Input/Output Unit	Process
Analog computer	Integrated Circuit (IC)	Program
Arithmetic Logic Unit (ALU)	Intelligent terminal	Punched cards
Assembly language	Jacquard's punch card	Second Generation
Babbage's Analytical Engine	Large Scale Integration (LSI)	Computer
Central Processing Unit (CPU)	Leibniz's Machine	Slide Rule
	Machine language	Smart phones
Computer	Mainframe computers	Software

Control Unit (CU)	Memory	Storage unit
Data	Microcomputers	Supercomputer
Desktop computer	Microprocessor	Super Large Scale Integrated (SLSI) chips
Digital computer	Minicomputers	Tablet computer
Dumb terminal	Napier's bones	Third Generation
Fifth Generation Computer	Netbook	Computer
First Generation Computer	Notebook computer	Transistors
Floating point Operations	Output	Users
Per Second (FLOPS)	Parallel processing	Vacuum Tubes
Fourth Generation Computer	Pascal's Machine	Very Large Scale Integration (VLSI)
Hardware	Personal Computer (PC)	
Hollerith's tabulator	Personal Digital Assistant (PDA)	
Input		

QUESTIONS

Section 1.2

1. Define an analog computer and a digital computer.
2. Give an example each of analog computer and digital computer.

Section 1.3

3. List the main characteristics of the computer.
4. Describe the characteristics of the computer.
5. List three significant limitations of the computer.

Section 1.4

6. Explain briefly the developments in computer technology starting from a simple calculating machine to the first computer.
7. What is a calculating machine?
8. What is the key feature of the Jacquard's punch card?
9. Name the first calculating device for the counting of large numbers.
10. Who is called the Father of Computer?

Section 1.5.1

11. The first generation computers used_____for circuitry.

12. Describe the first generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance, and (e) Their applications.

13. Give two examples of first generation computers.

14. List the drawbacks of the first generation computers.

Section 1.5.2

15. The second generation computers used_____for circuitry.

16. Describe the second generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance and (e) Their applications.

17. Give two examples of second generation computers.

18. List the drawbacks of the second generation computers.

Section 1.5.3

19. The third generation computers used_____for circuitry.

20. Describe the third generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance, and (e) Their applications.

21. Give two examples of third generation computers.

22. List the drawbacks of the third generation computers.

Section 1.5.4

23. The fourth generation computers used_____for circuitry.

24. Describe the fourth generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance and (e) Their applications.

25. Give two examples of fourth generation computers.

26. List the drawbacks of the fourth generation computers.

Section 1.5.5

27. The fifth generation computers used_____for circuitry.

28. Describe the fifth generation computer based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance and (e) Their applications.

29. Give two examples of fifth generation computers.

30. Compare in detail the five generations of computers based on the (a) Hardware (b) Software (c) Computing characteristics (d) Physical appearance and (e) Their applications. Also give at least one example of each generation of computer.

Section 1.6.1

31. Define microcomputer.

32. Give two examples of microcomputer.

33. List three categories of microcomputers.

Section 1.6.2

34. Define minicomputers.

35. Give two examples of minicomputer.

Section 1.6.3

36. Define mainframe computer.

37. Give two examples of mainframe computer.

38. Define a dumb terminal.

39. Define an intelligent terminal.

Section 1.6.4

40. Define a supercomputer.

41. Give two examples of supercomputer.

42. The speed of supercomputer is generally measured in_____.

43. List two uses of the supercomputer.

44. Name the supercomputer assembled in India.

45. Highlight the differences between microcomputer, minicomputer, mainframe computer and supercomputer.

Section 1.7

46. Define a computer.
47. Define (1) Program (2) Software (3) Hardware (4) ALU (5) CU (6) CPU (7) Data.
48. Differentiate between software, data and hardware.
49. List the components of computer hardware.
50. Explain in detail the components of computer hardware.
51. List the steps in the working of the computer.
52. Explain the working of the computer.
53. Explain the input-process-output cycle.

Section 1.8

54. List some areas where the computers are used.
55. Explain briefly the use of computers in the following areas—(a) Education, (b) Advertising, and (c) Government.

Extra Questions

56. Give full form of the following abbreviations

1. CPU
2. I/O
3. ALU
4. CU
5. LSI
6. VLSI
7. PC
8. GUI
9. SLSI
10. ES
11. NLP
12. AI
13. PDA
14. FLOPS
15. UNIVAC
16. ENIAC
17. EDVAC

57. Write short notes on

1. Components of Computer
2. Input-Process-Output
3. I/O Unit
4. Central Processing Unit
5. Storage Unit
6. History of Computers
7. First Generation Computer
8. Second Generation Computer
9. Third Generation Computer
10. Fourth Generation Computer
11. Fifth Generation Computer
12. Microcomputers
13. Minicomputers
14. Mainframe Computers
15. Supercomputer
16. Personal Computer (PC)
17. Notebook Computer
18. Tablet Computer
19. Netbook
20. Personal Digital Assistant (PDA)
21. Applications of Computer

58. Give differences between the following:

1. Analog and Digital Computer
2. Dumb Terminal and Intelligent Terminal
3. Microcomputer and Minicomputer
4. Minicomputer and Mainframe Computer
5. Mainframe computer and Supercomputer
6. First Generation Computers and Second Generation Computers
7. Second Generation Computers and Third Generation Computers
8. Third Generation Computers and Fourth Generation Computers
9. Fourth Generation Computers and Fifth Generation Computers
10. Desktop Computer and Notebook Computer